

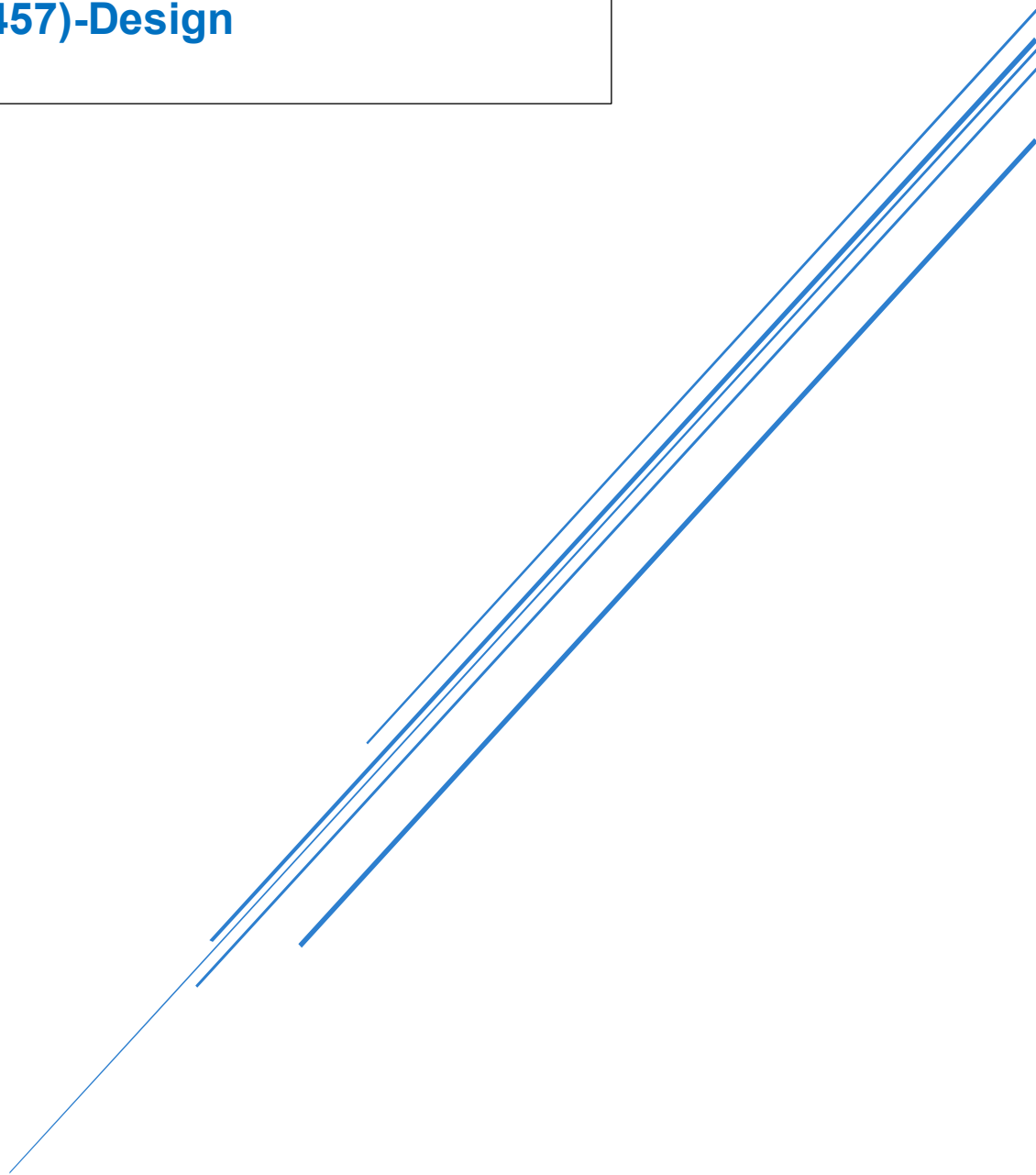
OPSC6311 PART 1

PLANNING & DESIGN

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**Oluga Neluvhalani (St10443034)-
Planning**

**Luthando Princess Mndawe
(St10446457)-Design**



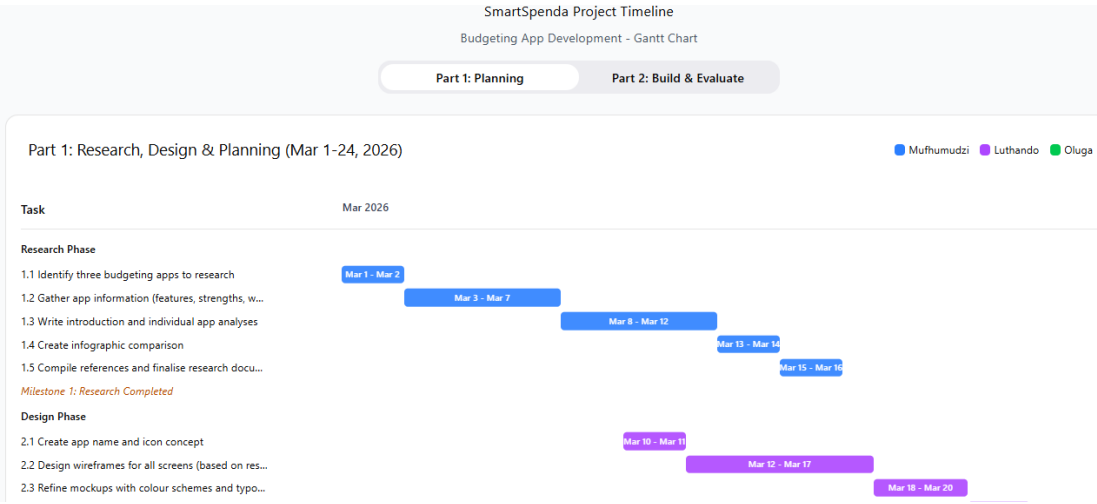
Introduction

This planning document serves as the roadmap for developing SmartSpenda which is a personal budget tracker application designed to make financial management engaging and rewarding. Following the comprehensive research conducted by my group member Mufhumudzi Rasilingwana (St10441516) on existing budgeting applications—Zaka Manager, Vault22, and Goodbudget—and the design work completed by Luthando Princess Mndawe (St10446457), I will now outline the structured approach to building this application.

The research phase revealed that successful budgeting apps share common elements: automated transaction tracking, strong privacy features, and engaging user interfaces. Zaka Manager demonstrated the value of SMS-based tracking without banking credentials, Vault22 showcased powerful automatic categorisation, and Goodbudget proved the effectiveness of envelope-style budgeting. Our design phase produced a clean, intuitive interface named "SmartSpenda" that combines these strengths with gamification elements to make budgeting feel less like a chore and more like a rewarding game.

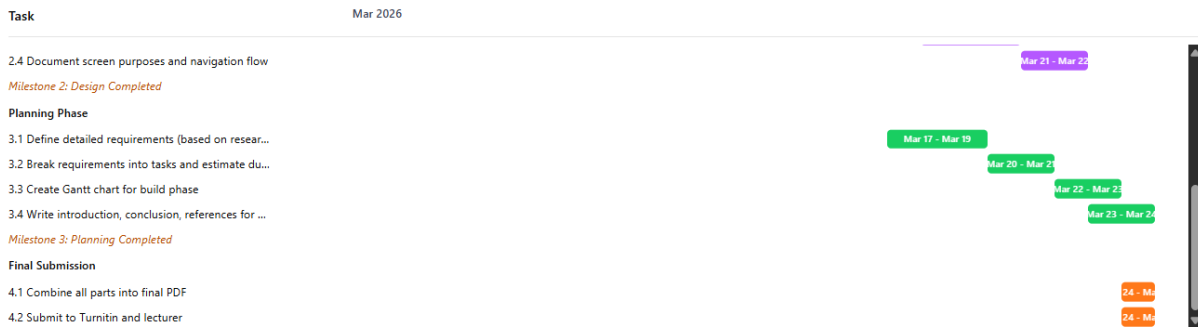
This planning document will outline the detailed requirements as well as provide a comprehensive project timeline through a Gantt chart, and establish clear milestones for development, testing, and final delivery.

Gantt Chart



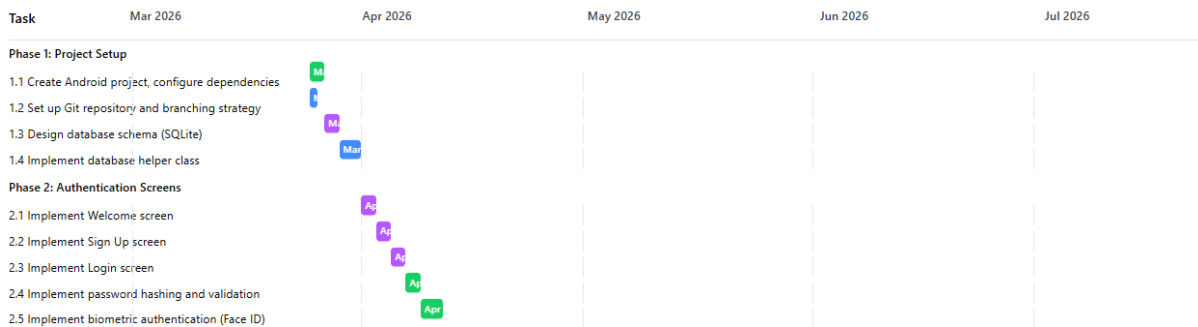
Part 1: Research, Design & Planning (Mar 1-24, 2026)

■ Mufhumudzi ■ Luthando ■ Oluga ■ All



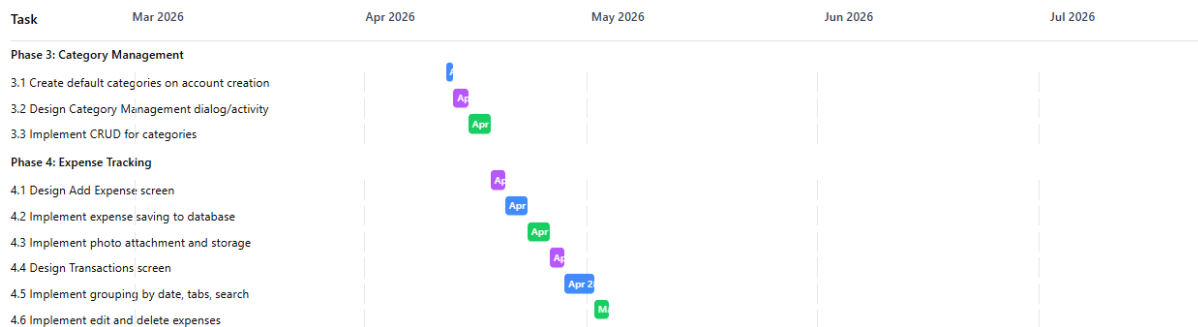
Part 2: Build & Evaluate (Mar 25 - Jul 26, 2026)

■ Mufhumudzi ■ Luthando ■ Oluga



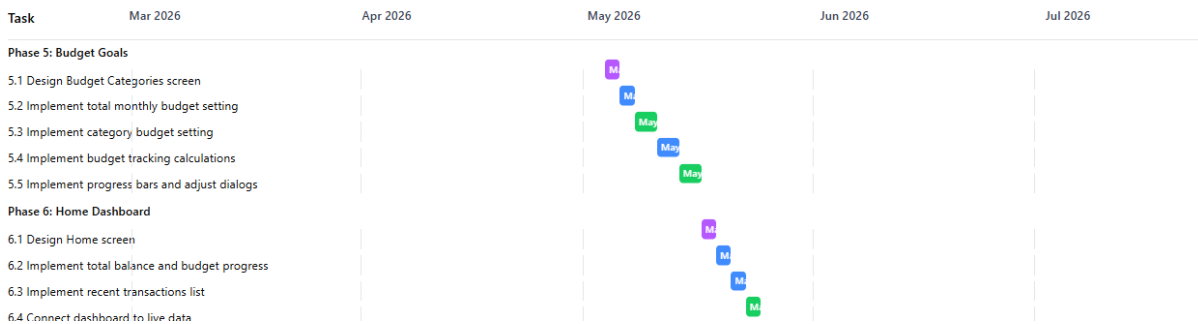
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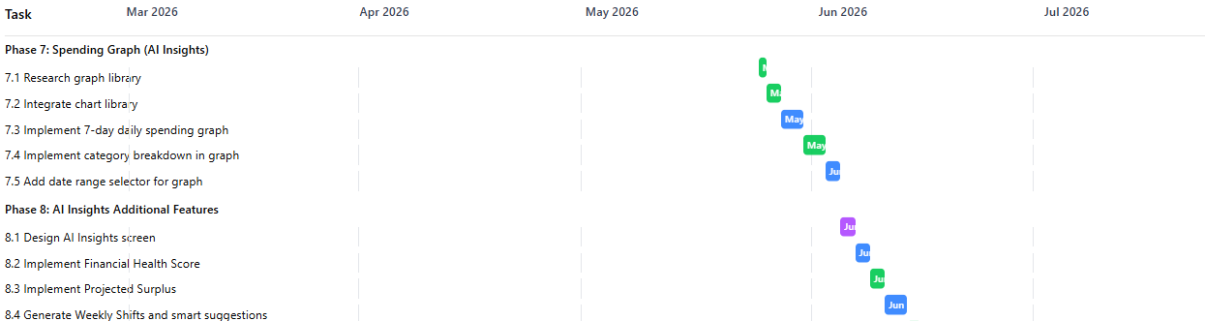
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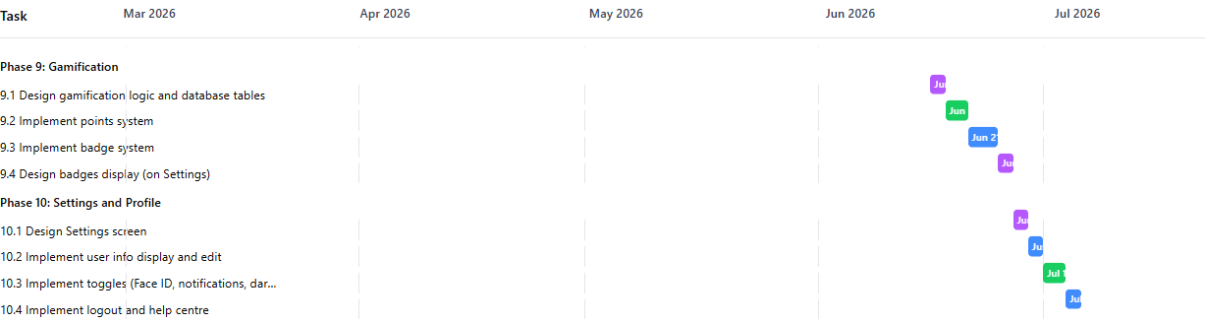
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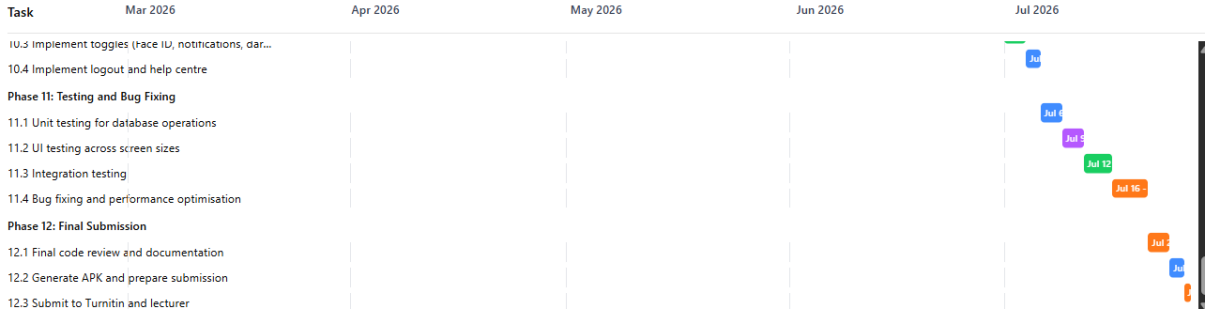
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Mufhumudzi Luthando Oluga All



Milestones

Part 1 Milestones (Completed)

Milestone	Date	Deliverable
Research Completed	2026-03-16	Research document (Zaka Manager, Vault22, Goodbudget analysis)
Design Completed	2026-03-22	Full set of screen mockups and navigation diagram
Planning Completed	2026-03-24	Planning document with requirements and Gantt charts

Part 1 Submitted	2026-03-24	Final PDF uploaded to Turnitin
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Part 2 Milestones (Planned)

Milestone	Date	Deliverable
M1: Project Initialisation	2026-03-31	Repository, database schema
M2: Authentication Complete	2026-04-11	Welcome, Sign Up, Login functional
M3: Core Expense Tracking	2026-05-03	Add, view, edit, delete expenses
M4: Budget Functionality	2026-05-16	Budget screen with limits and progress
M5: Dashboard Complete	2026-05-24	Home screen with real data
M6: Graph Integration	2026-06-04	7-day spending graph with date filter
M7: AI Insights Complete	2026-06-15	All insights elements working
M8: Gamification	2026-06-26	Points and badges system
M9: Settings Complete	2026-07-05	Profile and toggles
M10: Feature Complete	2026-07-05	All requirements implemented
M11: Testing Complete	2026	All tests passed, bugs fixed
M12: Final Submission	2026	APK and final POE submitted

App Overview & Designs: SmartSpenda

App Name and Branding

We are going to call this application **SmartSpenda** which is a name that combines "smart" financial management with the South African term "spenda" (spender), making it relatable to our target audience. The name suggests intelligent spending habits rather than restrictive budgeting.

App Icon Design

The icon will feature a stylised piggy bank with a subtle graph line integrated into its shape, using a gradient from deep teal to vibrant green. This colour scheme represents financial growth and stability while remaining visually appealing. A small coin with a ZAR "R" symbol will float above the piggy bank, clearly indicating South African Rand functionality. The icon will be recognisable at small sizes and will work well on both light and dark backgrounds.

Innovative Features

Drawing from the research we did SmartSpenda will incorporate the following innovative features:

1. **SMS-Based Transaction Detection (from Zaka Manager):** This will make users to be able to enable SMS reading to automatically detect and categorise expenses without sharing their banking credentials. This addresses the privacy concerns identified in Vault22 while maintaining its automation.
2. **Envelope Budgeting System (from Goodbudget):** This will make users to be able to allocate money to digital "envelopes" for different spending categories, helping them visualise exactly how much remains for each expense type.
3. **Gamification Elements:** This is our unique addition users earn "SmartPoints" for logging expenses, staying within budgets, and achieving savings goals. Points unlock badges, achievement trophies, and eventually premium features, transforming budgeting from a tedious task into an engaging experience "smart right".
4. **AI-Powered Insights (enhanced from Vault22):** The app will provide personalised spending insights and suggestions which will help users to understand their habits and make well informed decisions.

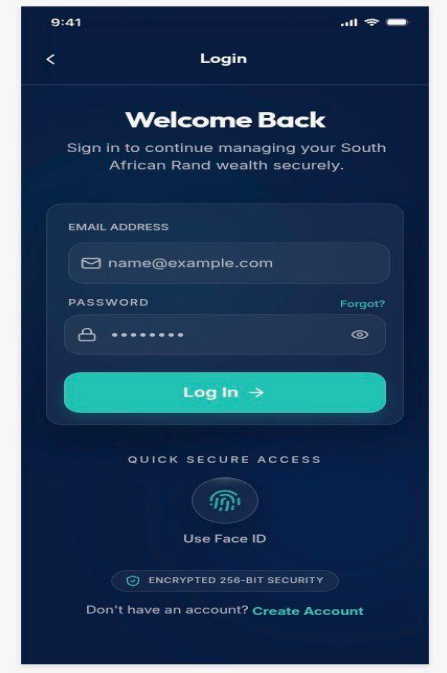
User Interface Design

The designs for all screens were created by Luthando Princess Mndawe (St10446457). The following screens and navigation structure will be implemented exactly as shown in the provided mockups. This section will outline every functional requirement for SmartSpenda which is structured to provide enough details for a developer to implement the application without needing additional clarification. Each requirement will be cross-referenced with the corresponding design screen.

1. User Authentication (Screens: Welcome, Login, Sign Up)

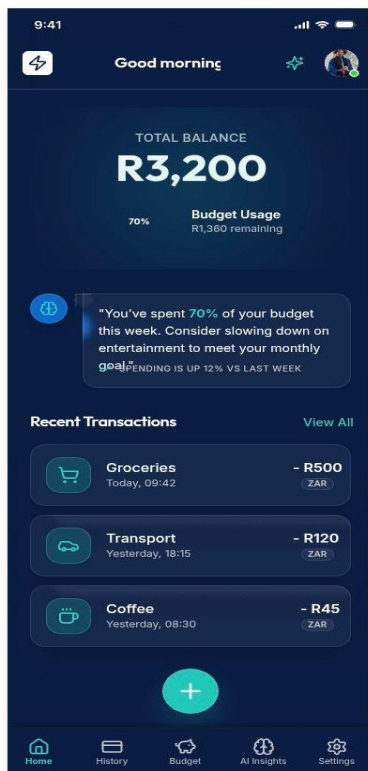
Requirement ID	Description
AUTH-01	Users must be able to register using an email address and password. Passwords must be at least 8 characters with one uppercase letter, one number, and one special character.
AUTH-02	Users must be able to log in using their registered credentials (as shown in the Login screen).
AUTH-03	The system must securely store passwords using hashing (bcrypt or similar).
AUTH-04	Biometric authentication (fingerprint/face ID) must be available as an optional login method for devices that support it (visible as "Use Face ID" on the Login screen).
AUTH-05	Users must be able to reset their password via "Forgot?" link on the Login screen.

AUTH-06	A welcome screen with “Sign Up” and “Log In” buttons must be shown on first launch (as per the “Smart spending starts here” design).
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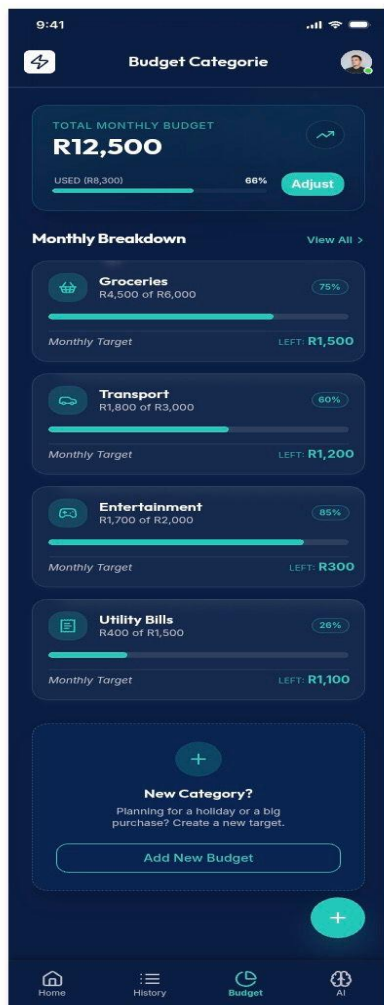
2. Progress Dashboard (Screen: Home/Dashboard)

Requirement ID	Description
DASH-01	The main dashboard must display the current month's budget usage (percentage and remaining amount).
DASH-02	The dashboard must show a visual representation (progress bar) of total budget usage.
DASH-03	Categories that are overspending must be visually highlighted in red (on the Budget Categories screen).
DASH-04	Categories nearing their limit (above 80%) must be visually highlighted in orange.
DASH-05	The dashboard must display recent transactions for quick access (as shown in the Home screen).



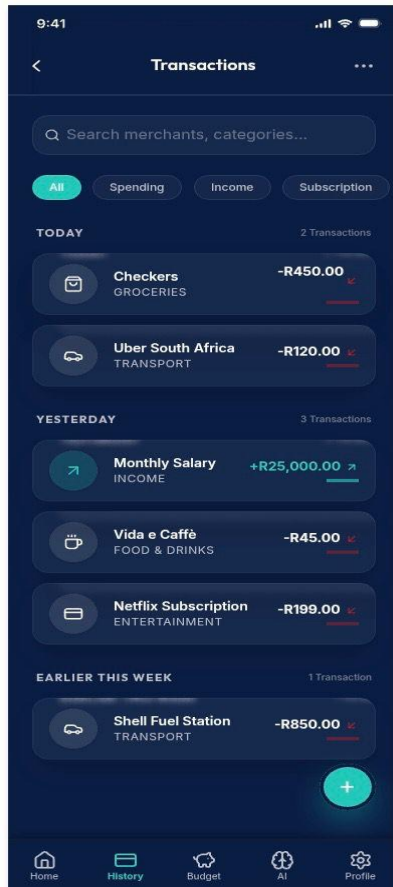
3. Budget Goals (Screen: Budget Categories)

Requirement ID	Description
BUD-01	Users must be able to set a monthly total budget goal (shown as "TOTAL MONTHLY BUDGET" at the top of the Budget Categories screen).
BUD-02	Users must be able to set individual budget limits for each expense category (displayed as e.g., "R6,000" for Groceries).
BUD-03	Budget limits must be able to be set for the current month and for future months.
BUD-04	The system must calculate and display the remaining budget for the current month (total and per category).
BUD-05	Users must be able to adjust budgets at any time (via the "Adjust" button next to the total budget and by editing category limits).



4. Expense List and Period Selection (Screen: Transactions)

Requirement ID	Description
LST-01	Users must be able to view a chronological list of all expenses (Transactions screen).
LST-02	Users must be able to filter expenses by a user-selectable date period (e.g., Today, Yesterday, Earlier this week) and by type (All, Spending, Income, Subscription).
LST-03	For each expense entry, the list must display: amount, description, category, and a thumbnail icon for any attached receipt photo.
LST-04	Tapping on an expense entry must open a detailed view showing all information including the full receipt photo if available.



5. AI Insights (Screen: AI Insights)

Requirement ID	Description
AI-01	The AI Insights screen must display a "Financial Health Score" (e.g., 84).
AI-02	It must show a "Projected Surplus" amount with an "Adjust Budget" button.
AI-03	It must show "Weekly Shifts" for key spending categories (e.g., Dining +20%).
AI-04	A smart suggestion must be displayed (e.g., "You spent 20% more on dining...").
AI-05	The 7-Day Spending Pattern graph must be interactive (tapping a bar shows details).
AI-06	"Top Category" and "Savings Progress" must be shown at the bottom.



6. Expense Entry (Screen: Add Expense)

Requirement ID	Description
EXP-01	Users must be able to add an expense entry with the following mandatory fields: amount (numeric, ZAR), date (date picker), description ("Notes"), category (selected from user's categories).
EXP-02	Users must be able to attach a photo receipt to any expense entry. The photo must be stored locally and associated with the expense record.
EXP-03	The system must validate that the amount is positive and that the date is not in the future.

9:41

Add Expense

ENTER AMOUNT

R
0.00

South African Rand (ZAR)

Select Category

Manage All

Groceries

Transport

Bills

Entertainment

Transaction Details

Date

2024-05-24

Notes (Optional)

e.g. Weekly organic vegetables

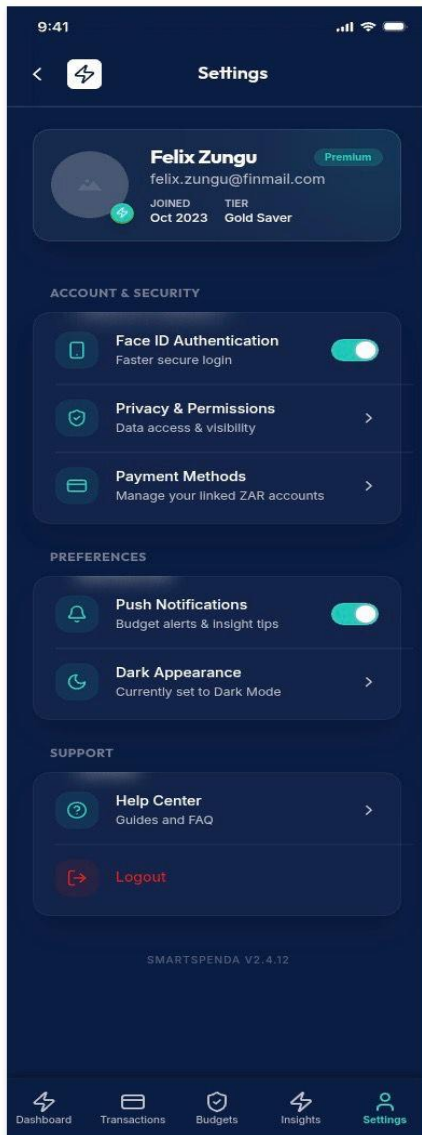
Budget Limit Alert

Adding this R0 to Groceries will bring you to 85% of your monthly budget.

+ Save Expense

7. Settings and Profile (Screen: Settings)

Requirement ID	Description
SET-01	Users must be able to view their profile information (name, email, join date, tier).
SET-02	Users must be able to toggle Face ID authentication.
SET-03	Users must be able to manage privacy permissions.
SET-04	Users must be able to toggle push notifications.
SET-05	Users must be able to switch between light and dark appearance (Dark Mode toggle).
SET-06	Users must be able to access a Help Center.
SET-07	Users must be able to log out.



Navigation Flow

The app uses a bottom navigation bar on the main screens (Home, History, Budget, AI Insights, Settings/Profile). The welcome screen leads to either Sign Up or Login. After login, the user lands on the Home screen.

Additional navigation patterns:

- Tapping the + (Floating Action Button) on the History screen opens the Add Expense screen.
- Tapping a transaction in the list opens a detailed view (or edit view).
- Tapping “Adjust” or editing category budgets on the Budget screen opens a dialog to modify limits.
- Tapping “Add New Budget” allows creating a new category with its own limit.

Conclusion

The planning phase for SmartSpenda has made us establish a clear and structured roadmap that bridges the research insights and the detailed UI designs created by the team. By analysing three existing budgeting applications Zaka Manager, Vault22, and Goodbudget we identified key features that balance automation, privacy, and user engagement. These findings directly informed the requirements and the visual direction that Luthando captured in the mockups, ensuring that every screen serves a purpose rooted in real user needs.

Our project plan, presented in Gantt chart format, breaks down the development process into manageable tasks with realistic timelines and assigned responsibilities. Milestones are set to track progress, and the schedule includes dedicated time for testing and refinement. This level of detail ensures that when we move into the build phase, we will have a shared understanding of what needs to be built and when.

With the research, design, and planning now complete, the team is fully prepared to begin implementing SmartSpenda. The planning document provides the foundation for a methodical, collaborative development effort that will ultimately deliver a functional, engaging, and well-tested personal budget tracker.

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